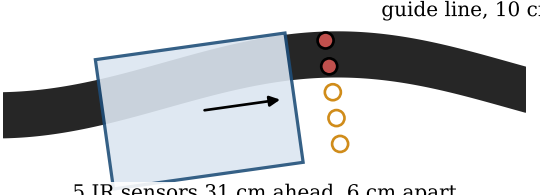
(a) BFS from hidden cells (0); descend; Chaikin ×3 = guide line

18	17	16	15	14	13	12	11	10	9	8	7	6	6
17	16	15	14	13	12	11	10	9		7	6	5	5
18	17	16	15	14		10	9	8		6	5	4	4
18	17	16	15	14		9	8	7	6	5	4	3	3
17	16	15	14	13		8	7	6	5	4	3	2	2
16	15	14	13	12		7	6					1	1
15	14	13	12	11		6	5	4	3	2	1	0	0
14	13	12	11	10		6	5	4	3	2	1	0	0
13	12	11	10	9	8	7	6	5	4	3	2	1	1

(b) PD line follower (top view, metres)



5 IR sensors 31 cm ahead, 6 cm apart

red = over the line: here $e = +0.75$ (turn left)

$e = \overline{\text{offset}}/0.12, \quad \omega = \text{clamp}(3.2e + 0.25\,\dot{e}, \pm 2.5)$

$v = 0.75(1 - 0.6|e|)$ m/s, wheels $v \mp \omega\, w/2$